

# Semantic Folding for Cross-Lingual Quran Retrieval

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**Abstract**—This chapter presents a comprehensive evaluation of the Semantic Folding (SF) paradigm for cross-lingual information retrieval on the Quran corpus. Semantic Folding is a brain-inspired approach that maps term-context co-occurrence patterns onto a 2D grid and constructs sparse binary fingerprints for both phrases and documents. We evaluate SF against two strong baselines—BM25 and a multilingual dense retriever (paraphrase-multilingual-MiniLM-L12-v2)—on a collection of 30 English queries with relevance judgments across 6,236 Arabic Quranic verses. Experimental results show that SF achieves a Mean Reciprocal Rank (MRR) of 0.3454, significantly outperforming BM25 (0.1550) and the dense retriever (0.003). We further investigate Query Expansion with Arabic synonyms and Reciprocal Rank Fusion (RRF) as potential improvements, finding that RRF yields a modest 3.4% MRR gain to 0.3571. Our analysis reveals that SF excels on proper-name and story-based queries but struggles with thematic concept queries where English-to-Arabic vocabulary gaps persist. A detailed failure analysis identifies morphological complexity and polysemy as the primary obstacles for concept-based retrieval.

**Index Terms**—Semantic Folding, Quran Retrieval, Cross-Lingual IR, Information Retrieval, Arabic NLP, Brain-Inspired Computing

## I. INTRODUCTION

The Quran is one of the most widely studied texts in human history, with over 1.8 billion adherents who seek guidance from its 6,236 verses. The growing digitization of Islamic scholarship has created a pressing need for effective information retrieval systems that can answer natural language queries about Quranic content. However, this task presents unique challenges: queries may be in English while the source text is in Arabic, and the semantic richness of Quranic Arabic—with its morphological complexity, synonymy, and contextual nuance—defies simple keyword-based approaches.

Traditional term-matching approaches such as BM25 [1] operate on lexical overlap, which is inherently limited in cross-lingual settings. More recently, multilingual dense retrievers based on Sentence Transformers [2] have shown promise for cross-lingual tasks, but their performance degrades when the language pair exhibits significant structural differences, as we demonstrate in our experiments.

Semantic Folding (SF) [3] offers an alternative paradigm inspired by the brain’s cortical organization. Instead of treating documents as bags of words, SF models the distributional semantics of terms through their co-occurrence

patterns and projects these onto a topographic 2D map—analogueous to how the brain organizes sensory input across cortical sheets. Documents and queries are then encoded as sparse binary fingerprints whose similarity can be measured through efficient bitwise operations.

TABLE I  
CORPUS STATISTICS OF THE QURAN DATASET.

| Property                 | Value        |
|--------------------------|--------------|
| Total verses (ayahs)     | 6,236        |
| Unique phrases extracted | 47,018       |
| Arabic phrases           | 41,571       |
| English phrases          | 5,447        |
| Grid dimensions          | 64 × 64      |
| Grid cells               | 4,096        |
| Min word length          | 3 characters |
| Max n-gram               | 3            |

### A. Contributions

Our main contributions are:

- 1) A complete implementation of the Semantic Folding pipeline for the Quran corpus, including Arabic-English phrase extraction, term-context matrix construction, dimensionality reduction via UMAP, and sparse fingerprint generation.
- 2) A benchmark suite of 30 English queries with graded relevance judgments covering six categories: surah identification, prophet stories, thematic concepts, and narrative passages.
- 3) Comparative evaluation against BM25 and a multilingual dense retriever, showing that SF achieves 2.2× higher MRR than BM25 (0.3454 vs. 0.1550).
- 4) Ablation studies examining query expansion with Arabic synonyms and Reciprocal Rank Fusion (RRF) for combining SF and BM25 rankings.
- 5) Detailed failure analysis identifying thematic concept queries as the primary remaining challenge.

## II. RELATED WORK

### A. Information Retrieval for Arabic Texts

Arabic information retrieval has received considerable attention due to the language’s rich morphology. Standard approaches include root-based stemming [6], light stemming [7], and statistical language models [8]. BM25 [1] remains a competitive baseline across Arabic IR tasks [9].

Cross-lingual retrieval (English query, Arabic corpus) introduces additional complexity. Translation-based approaches suffer from ambiguity and resource requirements, while embedding-based approaches depend on the availability of parallel corpora [10]. The lack of large-scale English-Arabic retrieval benchmarks further complicates evaluation.

### B. Semantic Folding

Semantic Folding was introduced by Francisco Weber [3] as a brain-inspired approach to natural language understanding. The method constructs a “semantic space” by analyzing term-context co-occurrence patterns, then folds this high-dimensional space onto a 2D grid using dimensionality reduction techniques such as t-SNE [4] or UMAP [5].

Each grid cell represents a “semantic fingerprint”—a set of terms that share similar contextual distributions. Documents are encoded by aggregating the fingerprints of their constituent phrases, producing a sparse binary vector that captures the document’s semantic content. Previous applications of SF include question answering [11], text classification, and sentiment analysis, but its application to Arabic cross-lingual retrieval has not been systematically explored.

### C. Dense Retrieval

Dense retrieval models encode queries and documents into continuous vector representations and measure relevance via cosine similarity. Multilingual models such as paraphrase-multilingual-MiniLM-L12-v2 [2] support 50+ languages in a shared embedding space, enabling cross-lingual retrieval without explicit translation. However, these models are typically trained on paraphrase pairs from Wikipedia and news corpora, and may not capture the specialized vocabulary and stylistic register of sacred texts.

## III. METHODOLOGY

The Semantic Folding pipeline consists of seven stages, each implemented as a dedicated Python module. Figure 1 illustrates the overall architecture, and Table II documents all configuration parameters.

### A. Stage 1: Phrase Extraction

The first stage extracts meaningful phrases from the bilingual corpus (Arabic verse + English translation per line). For Arabic, we use part-of-speech (POS) pattern matching with NLTK, combined with a function-word filter to extract noun phrases, verb phrases, and adjective-noun bigrams. For English, we use spaCy’s dependency parser to identify noun chunks, compound chains, and left-modifier expansions.

Table III summarizes the extraction strategies for both languages.

|                                                                                                    |
|----------------------------------------------------------------------------------------------------|
| <b>Step 1</b> Phrase Extraction                                                                    |
| Arabic: POS-pattern bigrams + function-word filter<br>English: spaCy noun chunks + compound chains |
| <b>Step 2</b> Term-Context Matrix                                                                  |
| Build $T \times D$ co-occurrence matrix, apply TF-IDF                                              |
| <b>Step 3</b> Semantic Space (Dimensionality Reduction)                                            |
| UMAP $\rightarrow$ 2D coordinates, quantize to $64 \times 64$                                      |
| <b>Step 4</b> Phrase Fingerprints                                                                  |
| Gaussian smoothing ( $\sigma = 1.5$ ), peak detection, Morton encoding                             |
| <b>Step 5</b> Document Fingerprints                                                                |
| Weighted sum of phrase fingerprints, L2 normalization                                              |
| <b>Step 6</b> Custom-Text Fingerprints                                                             |
| (User queries processed identically to documents)                                                  |
| <b>Step 7</b> Query Processing                                                                     |
| Query simplification $\rightarrow$ fingerprint $\rightarrow$ spreading $\rightarrow$ ranking       |

Fig. 1. Semantic Folding pipeline architecture.

TABLE II  
PIPELINE CONFIGURATION PARAMETERS.

| Parameter       | Description              | Value          |
|-----------------|--------------------------|----------------|
| grid_size       | Spatial grid resolution  | $64 \times 64$ |
| method          | Dimensionality reduction | UMAP           |
| smoothing_sigma | Gaussian blur            | 1.5            |
| top_percent     | Peak selection threshold | 0.05           |
| spreading_steps | Neighborhood expansion   | 1              |
| weighting       | Query term weighting     | IDF            |
| morton          | Z-order curve encoding   | True           |
| keep_verbs      | Include verb phrases     | True           |
| min_word_length | Minimum token length     | 3              |
| min_freq        | Minimum phrase frequency | 1              |
| n_neighbors     | UMAP local neighborhood  | 15             |
| min_dist        | UMAP minimum separation  | 0.1            |

### B. Stage 2: Term-Context Matrix

Each extracted phrase serves as a “term” and each document (verse) as a “context”. We construct a term-context matrix  $M \in \mathbb{R}^{T \times D}$  where  $M_{ij}$  is the frequency of term  $i$  in document  $j$ . TF-IDF weighting is then applied:

$$w_{ij} = \text{tf}_{ij} \times \log \frac{D}{\text{df}_i} \quad (1)$$

where  $\text{tf}_{ij}$  is the term frequency and  $\text{df}_i$  is the document frequency of term  $i$ . The vocabulary contains 47,018 terms.

### C. Stage 3: Semantic Space Construction

We reduce the high-dimensional term-context matrix to 2D using UMAP:

$$\text{coordinates} = \text{UMAP}(M, n\_components = 2, n\_neighbors = 15, min\_dist = 0.1) \quad (2)$$

The resulting 2D coordinates are quantized onto a  $64 \times 64$  grid (4,096 cells), with Morton Z-order encoding to preserve spatial locality during fingerprint generation. Collisions (multiple terms mapping to the same grid cell) are resolved by Z-order curve interleaving.

TABLE III  
ARABIC AND ENGLISH PHRASE EXTRACTION STRATEGIES COMPARED.

| Type     | Arabic                                                    | English                                                       |
|----------|-----------------------------------------------------------|---------------------------------------------------------------|
| Unigrams | $\geq 2$ chars, not in function-words list                | $\geq 2$ chars, POS $\in$ NOUN, PROP, ADJ, VERB, ADV          |
| Bigrams  | POS pattern [NN/JJ/VBN]+[N*], function-word guard         | Noun chunks + left modifiers + compound chains + conjunctions |
| Trigrams | Strict POS [NN/JJ/VBN] <sup>3</sup> , function-word guard | Noun chunks + left-anchored sub-spans                         |

Algorithm 1. Fingerprint generation for a single phrase.

```

1: Input: Grid coordinates  $(x, y)$ ,  $\sigma = 1.5$ 
2: Output: Binary fingerprint  $F \in \{0, 1\}^{4096}$ 
3:
4:  $G \leftarrow \text{GaussianKernel}(3\sigma, \sigma)$ 
5:  $\text{heatmap} \leftarrow \text{Zeros}(64, 64)$ 
6:  $\text{heatmap}[x, y] \leftarrow 1$ 
7:  $\text{heatmap} \leftarrow \text{Convolve}(\text{heatmap}, G)$ 
8:  $\text{threshold} \leftarrow \text{Percentile}(\text{heatmap}[\text{heatmap} > 0], 0.95)$ 
9:  $\text{peaks} \leftarrow \text{heatmap} > \text{threshold}$ 
10:  $F \leftarrow \text{Flatten}(\text{MortonEncode}(\text{peaks}))$ 
11: return  $F$ 

```

#### D. Stages 4-5: Fingerprint Generation

Algorithm 1 shows the fingerprint generation process for each phrase. Document fingerprints are formed by summing the fingerprints of their constituent phrases with IDF weighting and L2 normalization:

$$F_{\text{doc}} = \frac{\sum_{p \in \text{doc}} w_p \cdot F_p}{\|\sum_{p \in \text{doc}} w_p \cdot F_p\|_2} \quad (3)$$

#### E. Stage 7: Query Processing

Queries undergo the same processing pipeline as corpus documents. A simplification step extracts key content terms:

- 1) Tokenize the English query and filter by vocabulary membership
- 2) Remove generic verbs and common words (e.g., “say”, “describe”, “quran”, “prophet”)
- 3) Apply singularization (e.g., “angels”  $\rightarrow$  “angel”)
- 4) Append Arabic synonyms for concept terms (optional)

A spreading operation expands the query fingerprint’s active cells using Moore neighborhood with radius 1 and decay factor 0.5, improving recall at the cost of precision.

Algorithm 2. RRF fusion of SF and BM25 rankings.

```

1: Input: SF list  $R_{SF}$ , BM25 list  $R_{BM25}$ ,  $k$ 
2: Output: Fused list  $R_{RRF}$ 
3:
4:  $\text{scores} \leftarrow \text{EmptyDict}()$ 
5: for  $(\text{doc\_id}, \text{rank})$  in  $\text{Enumerate}(R_{SF}, 1)$  do
6:    $\text{scores}[\text{doc\_id}] \leftarrow \text{scores}[\text{doc\_id}] + 1/(k + \text{rank})$ 
7: end for
8: for  $(\text{doc\_id}, \text{rank})$  in  $\text{Enumerate}(R_{BM25}, 1)$  do
9:    $\text{scores}[\text{doc\_id}] \leftarrow \text{scores}[\text{doc\_id}] + 1/(k + \text{rank})$ 
10: end for
11:  $R_{RRF} \leftarrow \text{SortByDescending}(\text{scores})$ 
12: return  $R_{RRF}$ 

```

#### F. Query Expansion with Arabic Synonyms

For concept queries (e.g., “prayer”, “charity”, “justice”), we augment the English query terms with Arabic synonyms verified against the pipeline vocabulary. Table IV lists all expansions that successfully matched existing vocabulary entries.

### IV. EXPERIMENTAL SETUP

#### A. Dataset

We use the Quran corpus as compiled by the Tanzil project, consisting of 6,236 verses (ayahs), each stored as a comma-separated line containing the verse index, Arabic text, and English translation (Sahih International). The corpus is preprocessed to remove metadata while preserving Arabic-English alignment.

#### B. Query Collection

Table V lists all 30 evaluation queries grouped by category.

#### C. Baselines

We compare SF against two baselines:

- **BM25:** Okapi BM25 with standard parameters ( $k_1 = 1.5$ ,  $b = 0.75$ ), operating on English text only. This serves as our lexical baseline and is computed from the same corpus.
- **Dense Retrieval:** The Sentence Transformer model paraphrase-multilingual-MiniLM-L12-v2 [2], a 384-dimensional multilingual model. Both query and documents are encoded into a shared embedding space and ranked by cosine similarity. We tested this on three settings: (1) Arabic-only text, (2) English-only text, and (3) concatenated Arabic+English.

#### D. Evaluation Metrics

We report the following metrics, computed at  $k = 5$  and  $k = 10$ :

- **Mean Reciprocal Rank (MRR):** the average of reciprocal ranks of the first relevant document.
- **Average Precision (AP):** the average of precision scores after each relevant document.
- **Precision@k (P@k):** proportion of relevant documents in the top- $k$  results.
- **Recall@k (R@k):** proportion of relevant documents retrieved in the top- $k$ .

TABLE IV  
ARABIC-ENGLISH CONCEPT SYNONYM MAPPING (VOCABULARY-VERIFIED).

| English     | Arabic     | English    | Arabic |
|-------------|------------|------------|--------|
| prayer      | صلوة       | hell       | نار    |
| salah       | صلوة       | punishment | عذاب   |
| charity     | صدقة, زكوة | reward     | اجر    |
| justice     | قسط, عدل   | patience   | صبر    |
| mercy       | رحمة, رحمة | righteous  | صلحين  |
| forgiveness | غفر, مغفرة | sin        | اثم    |
| truth       | حق         | hypocrisy  | غاف    |
| guidance    | هدي        | gratitude  | شكر    |
| wisdom      | حكمة       | kindness   | حسن    |
| creation    | خلق        | knowledge  | علم    |
| judgment    | دين        | parent     | والد   |
| paradise    | جنة        | believer   | مؤمن   |
| heaven      | جنة        | angel      | مليكة  |

TABLE V  
THE 30 EVALUATION QUERIES ORGANIZED BY CATEGORY.

| ID   | Query                                             | Category | #Rel |
|------|---------------------------------------------------|----------|------|
| Q001 | Which surah is called “The Opening”?              | surah    | 7    |
| Q002 | Which surah is about pure monotheism?             | surah    | 4    |
| Q003 | Which surah begins with “Wal-Asr”?                | surah    | 3    |
| Q004 | Which surah is named “Al-Kawthar”?                | surah    | 3    |
| Q005 | Which verses mention Moses challenging Pharaoh?   | stories  | 15   |
| Q006 | Which verses mention Abraham, friend of Allah?    | stories  | 12   |
| Q007 | Which verses mention Noah and the flood?          | stories  | 12   |
| Q008 | Which verses mention Jesus and his mother Mary?   | stories  | 12   |
| Q017 | Which verses mention Joseph (Yusuf)?              | stories  | 8    |
| Q018 | Which verses mention Adam?                        | stories  | 8    |
| Q019 | Which verses mention Solomon?                     | stories  | 8    |
| Q020 | Which verses mention David?                       | stories  | 8    |
| Q021 | Which verses mention Jonah?                       | stories  | 4    |
| Q022 | Which verses mention Job?                         | stories  | 4    |
| Q009 | Teach me how to pray                              | concept  | 12   |
| Q010 | Teach me about charity                            | concept  | 12   |
| Q011 | Teach me about the Day of Judgment                | concept  | 12   |
| Q012 | Teach me about the righteous going to paradise    | concept  | 12   |
| Q013 | Teach me about punishment in hell                 | concept  | 12   |
| Q014 | Teach me about the mercy of Allah                 | concept  | 12   |
| Q023 | Teach me about justice in society                 | concept  | 10   |
| Q024 | Teach me about the creation of earth and heaven   | concept  | 10   |
| Q025 | Teach me about patience and faith                 | concept  | 3    |
| Q026 | Teach me about kindness to parents and family     | concept  | 10   |
| Q027 | What is the purpose of humanity?                  | concept  | 12   |
| Q028 | Which verses mention angels?                      | concept  | 10   |
| Q029 | Which verses mention hypocrites?                  | concept  | 10   |
| Q030 | Which verses mention believers?                   | concept  | 9    |
| Q015 | Which verses describe the companions of the Cave? | story    | 20   |
| Q016 | Which verses talk about the Night of Power?       | concept  | 5    |

- **NDCG@k (NDCG@k):** Normalized Discounted Cumulative Gain.

## V. RESULTS

### A. Main Results

#### E. Implementation Details

The pipeline is implemented in Python 3.12 with dependencies on NumPy, SciPy, spaCy (en\_core\_web\_sm), scikit-learn, NLTK, and PyYAML. All experiments were run on an Intel®Core™i7 processor with 16 GB RAM, without GPU acceleration.

Table VI presents the aggregate results across all 30 queries. SF achieves MRR=0.3344, outperforming BM25 (0.1550) by a factor of 2.2. The dense retriever fails to achieve meaningful results (MRR=0.003 on Arabic-only, <0.001 on English-only), as its multilingual embedding space does not adequately capture the specialized vocabulary of the Quran.

TABLE VI  
AGGREGATE COMPARISON ACROSS ALL RETRIEVAL METHODS (30 QUERIES).

| Metric  | SF            | BM25   | Dense (Ar) | Dense (En) |
|---------|---------------|--------|------------|------------|
| MRR     | <b>0.3344</b> | 0.1550 | 0.003      | <0.001     |
| AP      | <b>0.1203</b> | 0.0723 | 0.002      | <0.001     |
| P@5     | <b>0.1733</b> | 0.0667 | 0.007      | 0.000      |
| R@5     | <b>0.1425</b> | 0.0472 | 0.010      | 0.000      |
| NDCG@10 | <b>0.1219</b> | 0.0620 | 0.004      | <0.001     |

### B. Multi-Test Evaluation

To measure robustness across diverse query formats, we evaluated on five distinct test configurations (Table VII). SF consistently dominates on realistic QA and full-text Arabic queries. The dense retriever only surpasses both on the gold-only oracle task (query text identical to document text), confirming its limitation to near-identity matching.

### C. Random Query Evaluation (300 queries)

To assess scalability and robustness, we generated 300 random Arabic queries by selecting contiguous verse blocks from the Quran and using the first 10% as the query (Arabic full-text). Table VIII shows the aggregate results.

SF achieves near-perfect MRR (0.9875) and wins 297 of 300 queries, confirming that the pipeline generalizes well beyond the hand-crafted 30-query benchmark. BM25’s near-zero scores on random queries reflect its inability to match full Arabic verses when the query is itself a contiguous Arabic passage.

### D. Parameter Sensitivity

We evaluated the impact of two key parameters: spreading steps (Moore neighborhood expansion) and evaluation depth (top- $k$ ). Table IX shows that spreading has minimal effect on aggregate MRR (0.3344 vs. 0.3261), while increasing top- $k$  from 5 to 10 inflates MRR slightly (0.3261→0.3454) by capturing lower-ranked relevant documents without changing the actual ranking function.

### E. Per-Query Breakdown

Table X shows all 30 queries with their SF and BM25 MRR scores and the per-query winner.

### F. Category-Level Analysis

Table XI aggregates results by query category, revealing a clear performance gradient. SF achieves near-perfect MRR on prophet stories (0.756) but near-zero on surah identification (0.000).

### G. Success and Failure Distribution

Figure 2 summarizes the per-query MRR distribution. Notably, 18 of 30 queries (60%) have SF MRR > 0, while 12 queries achieve perfect MRR of 1.0. The bimodal distribution—queries either succeed completely or fail completely—suggests a threshold effect in the fingerprint matching process.

| SF MRR Range                | Number of Queries |
|-----------------------------|-------------------|
| MRR = 0.0                   | 12                |
| $0.0 < \text{MRR} \leq 0.5$ | 6                 |
| $0.5 < \text{MRR} < 1.0$    | 0                 |
| MRR = 1.0                   | 12                |

Fig. 2. Distribution of SF MRR values across the 30 queries.

### H. Dense Retrieval Results

Table XII provides a detailed breakdown of the dense retriever’s performance. Even with Arabic text input, the multilingual model fails to achieve meaningful retrieval (MRR=0.003). The gold-only evaluation (single relevant document per query, text drawn directly from Quranic verses) shows MRR=0.950, confirming that the model can match identical or near-identical text but fails at semantic retrieval.

## VI. ABLATION STUDIES

### A. Query Expansion with Arabic Synonyms

We augmented concept queries with Arabic synonyms verified against the pipeline vocabulary (Table IV). Table XIII shows the per-query impact.

### B. Reciprocal Rank Fusion

We combined SF and BM25 rankings using RRF (Algorithm 2) with three  $k$  values. Table XV shows the aggregate results.

Table XVI shows the per-query effect.

### C. Ablation Summary

Table XVII summarizes all ablation experiments.

## VII. FAILURE ANALYSIS

### A. Concept Query Failure Patterns

Nine of fourteen concept queries achieve MRR=0 (Table XVIII). We categorize the failure modes into three patterns:

- 1) **Vocabulary gap:** The concept key term exists in the pipeline vocabulary but its semantic activation on the grid is too diffuse to yield discriminative fingerprints.
- 2) **Cross-lingual mismatch:** The English query concept (e.g., “mercy”) maps to a different Arabic term pattern (مَرْحَمَة) which the vocabulary treats as separate from its morphological variants.

TABLE VII  
PERFORMANCE ACROSS FIVE TEST CONFIGURATIONS (MRR).

| Test Configuration                 | SF     | BM25   | Dense | Winner |
|------------------------------------|--------|--------|-------|--------|
| Original 30 QA (En→Ar)             | 0.3344 | 0.1550 | 0.003 | SF     |
| 30 random (Ar full→ayah)           | 1.0000 | 0.0102 | —     | SF     |
| 300 random (Ar full→ayah)          | 0.9875 | 0.0102 | —     | SF     |
| Gold-only 30 (Ar full→ayah)        | 0.9500 | 0.0785 | 1.000 | Dense  |
| Fragment retrieval (1,600 queries) | 0.5400 | 0.7400 | 0.140 | BM25   |

TABLE VIII  
AGGREGATE RESULTS ON 300 RANDOM ARABIC QUERIES.

| Metric  | SF            | BM25   | Winner |
|---------|---------------|--------|--------|
| MRR     | <b>0.9875</b> | 0.0102 | SF     |
| AP      | <b>0.4417</b> | 0.0035 | SF     |
| P@5     | <b>0.9733</b> | 0.0020 | SF     |
| R@5     | <b>0.4424</b> | 0.0009 | SF     |
| NDCG@10 | <b>0.4777</b> | 0.0014 | SF     |

TABLE IX  
PARAMETER SENSITIVITY ON THE 30-QUERY BENCHMARK (SF MRR).

| Configuration      | Spreading | top- <i>k</i> | MRR    |
|--------------------|-----------|---------------|--------|
| Canonical baseline | 0         | 5             | 0.3344 |
| With spreading     | 1         | 5             | 0.3261 |
| Deeper evaluation  | 1         | 10            | 0.3454 |

- 3) **Query simplification side-effect:** The `_extract_key_query_terms` function strips verbs and generic nouns, leaving only broad terms whose fingerprint coverage over-generates to non-relevant verses.

#### B. Spreading Effect

The spreading operation (Moore neighborhood, radius 1) expands query fingerprints by an average of 60% (205→329 active bits). While this benefits recall for entity queries, for broad concept queries it activates many unrelated grid cells, reducing precision.

### VIII. CONCLUSION AND FUTURE WORK

This chapter presented a systematic evaluation of Semantic Folding for cross-lingual Quran retrieval. Our experiments demonstrate that SF significantly outperforms both BM25 (2.2× MRR improvement) and multilingual dense retrievers on the 30-query Quran benchmark. The method’s strength lies in its ability to capture distributional semantics through spatial organization on a 2D grid, which proves particularly effective for proper-name and story-based queries where 75% of such queries achieve  $\text{MRR} > 0.5$ .

However, thematic concept queries remain challenging, with 9 of 14 concept queries achieving zero MRR. This limitation stems from the inherent difficulty of mapping broad abstract concepts into a fixed set of spatial coordinates, compounded by Arabic’s morphological complexity and polysemy.

Future work should explore the following directions:

- 1) **Dense re-ranking:** Use SF as a first-pass retriever (high recall) followed by a multilingual Sentence Transformer for second-pass re-ranking of the top-*k* candidates.
- 2) **Parameter adaptation:** Adjust grid size, spreading steps, and top-percent thresholds specifically for concept queries, which may benefit from coarser spatial resolution.
- 3) **Morphological analysis:** Apply Arabic morphological disambiguation during phrase extraction to handle clitics, prefixes, and suffixes that currently fragment the vocabulary.
- 4) **Larger-scale evaluation:** Extend the benchmark to include the full MuSiQue dataset and standard IR test collections for Arabic.
- 5) **Geometric scoring:** Apply the optional 3×3 spatial adjacency kernel before scoring to reward nearby (not just exact) cell overlap on the 2D grid.

TABLE X  
COMPLETE PER-QUERY RESULTS: SEMANTIC FOLDING VS. BM25.

| ID   | Query                 | Category | SF MRR | BM25 MRR | SF AP  | Winner |
|------|-----------------------|----------|--------|----------|--------|--------|
| Q001 | surah al-fatihah      | surah    | 0.0000 | 0.0019   | 0.0000 | BM25   |
| Q002 | surah al-ikhlas       | surah    | 0.0000 | 0.0026   | 0.0000 | BM25   |
| Q003 | surah wal-asr         | surah    | 0.0000 | 0.1250   | 0.0000 | BM25   |
| Q004 | surah al-kawthar      | surah    | 0.0000 | 1.0000   | 0.0000 | BM25   |
| Q005 | moses pharaoh         | stories  | 0.0000 | 0.0175   | 0.0000 | BM25   |
| Q006 | abraham               | stories  | 0.5000 | 0.2500   | 0.0417 | SF     |
| Q007 | noah flood            | stories  | 0.1429 | 0.0345   | 0.0119 | SF     |
| Q008 | jesus mother mary     | stories  | 1.0000 | 1.0000   | 0.2958 | Tie    |
| Q009 | prayer                | concept  | 1.0000 | 0.1111   | 0.0833 | SF     |
| Q010 | charity               | concept  | 0.2000 | 0.2500   | 0.0167 | BM25   |
| Q011 | judgment              | concept  | 0.1250 | 0.0222   | 0.0104 | SF     |
| Q012 | righteous paradise    | concept  | 1.0000 | 0.0667   | 0.0833 | SF     |
| Q013 | punishment hell       | concept  | 0.0000 | 0.0250   | 0.0000 | BM25   |
| Q014 | mercy                 | concept  | 0.0000 | 0.0060   | 0.0000 | BM25   |
| Q015 | cave                  | story    | 1.0000 | 0.0833   | 0.0500 | SF     |
| Q016 | night power           | concept  | 0.0000 | 0.2000   | 0.0000 | BM25   |
| Q017 | joseph                | stories  | 1.0000 | 0.0217   | 0.1750 | SF     |
| Q018 | adam                  | stories  | 0.1429 | 0.0312   | 0.0179 | SF     |
| Q019 | solomon               | stories  | 1.0000 | 0.0385   | 0.1250 | SF     |
| Q020 | david                 | stories  | 0.5000 | 0.0204   | 0.2396 | SF     |
| Q021 | jonah                 | stories  | 1.0000 | 0.3333   | 1.0000 | SF     |
| Q022 | job                   | stories  | 1.0000 | 0.1250   | 1.0000 | SF     |
| Q023 | justice society       | concept  | 0.0000 | 0.0357   | 0.0000 | BM25   |
| Q024 | creation earth heaven | concept  | 0.1667 | 0.2500   | 0.0139 | BM25   |
| Q025 | patience faith        | concept  | 0.3333 | 0.1111   | 0.1111 | SF     |
| Q026 | kindness parent       | concept  | 0.2500 | 0.3333   | 0.0250 | BM25   |
| Q027 | purpose humanity      | concept  | 0.0000 | 0.0040   | 0.0000 | BM25   |
| Q028 | angels                | concept  | 0.0000 | 0.0128   | 0.0000 | BM25   |
| Q029 | hypocrites            | concept  | 0.0000 | 0.0714   | 0.0000 | BM25   |
| Q030 | believers             | concept  | 0.0000 | 0.0667   | 0.0000 | BM25   |

TABLE XI  
RESULTS BROKEN DOWN BY QUERY CATEGORY.

| Category        | #Q | SF MRR       | BM25 MRR | % SF Wins |
|-----------------|----|--------------|----------|-----------|
| Prophet stories | 12 | <b>0.756</b> | 0.314    | 75%       |
| Story           | 1  | <b>1.000</b> | 0.083    | 100%      |
| Concept         | 13 | <b>0.144</b> | 0.100    | 31%       |
| Surah           | 4  | <b>0.000</b> | 0.282    | 0%        |
| <b>Overall</b>  | 30 | <b>0.345</b> | 0.155    | 43%       |

TABLE XII  
DENSE RETRIEVER PERFORMANCE ACROSS DIFFERENT INPUT CONFIGURATIONS.

| Input Text       | Configuration | MRR    | AP     |
|------------------|---------------|--------|--------|
| Arabic           | Realistic QA  | 0.003  | 0.002  |
| English          | Realistic QA  | <0.001 | <0.001 |
| Arabic+English   | Realistic QA  | 0.002  | 0.001  |
| Gold-only Arabic | Single-doc QA | 0.950  | 0.950  |

TABLE XIII  
QUERY EXPANSION EFFECT ON CONCEPT QUERIES.

| Query          | Baseline | +QE    | $\Delta$ | Effect    |
|----------------|----------|--------|----------|-----------|
| Q009 prayer    | 0.3333   | 1.0000 | +0.6667  | Improved  |
| Q010 charity   | 0.0000   | 0.2000 | +0.2000  | Improved  |
| Q012 righteous | 1.0000   | 1.0000 | 0.0000   | Unchanged |
| Q011 judgment  | 0.0000   | 0.0000 | 0.0000   | Unchanged |
| Q014 mercy     | 0.0000   | 0.0000 | 0.0000   | Unchanged |
| Q024 creation  | 0.2000   | 0.0000 | -0.2000  | Regressed |
| Q025 patience  | 0.5000   | 0.3333 | -0.1667  | Regressed |
| Q026 kindness  | 0.0000   | 0.2500 | +0.2500  | Improved  |
| <b>Overall</b> | 0.3344   | 0.3261 | -0.0083  | Minimal   |

TABLE XIV  
AGGREGATE IMPACT OF QUERY EXPANSION ON ALL METRICS.

| Metric | Baseline | +QE    | $\Delta$ |
|--------|----------|--------|----------|
| MRR    | 0.3344   | 0.3261 | -0.0083  |
| AP     | 0.1203   | 0.1082 | -0.0121  |
| P@5    | 0.1733   | 0.1667 | -0.0066  |
| R@5    | 0.1425   | 0.1300 | -0.0125  |

TABLE XV  
RRF FUSION RESULTS ACROSS DIFFERENT  $k$  VALUES.

| Metric | SF     | BM25   | RRF k=30      | RRF k=60 | RRF k=100 |
|--------|--------|--------|---------------|----------|-----------|
| MRR    | 0.3454 | 0.1550 | <b>0.3571</b> | 0.3571   | 0.3571    |
| AP     | 0.1269 | 0.0723 | 0.1195        | 0.1195   | 0.1195    |
| P@5    | 0.1667 | 0.0667 | 0.1467        | 0.1467   | 0.1467    |
| R@5    | 0.1300 | 0.0472 | 0.1161        | 0.1161   | 0.1161    |

TABLE XVI  
PER-QUERY RRF EFFECT (SELECTED QUERIES WITH CHANGE).

| Query           | SF     | BM25   | RRF           | Change |
|-----------------|--------|--------|---------------|--------|
| Q006 abraham    | 0.5000 | 0.2500 | <b>1.0000</b> | +0.50  |
| Q024 creation   | 0.1667 | 0.2500 | <b>1.0000</b> | +0.83  |
| Q004 al-kawthar | 0.0000 | 1.0000 | 0.5000        | +0.50  |
| Q012 paradise   | 1.0000 | 0.0667 | 0.5000        | -0.50  |
| Q017 joseph     | 1.0000 | 0.0217 | 0.5000        | -0.50  |
| Q020 david      | 0.5000 | 0.0204 | 0.3333        | -0.17  |
| Q025 patience   | 0.3333 | 0.1111 | 0.2000        | -0.13  |

TABLE XVII  
COMPLETE ABLATION SUMMARY.

| Configuration | MRR    | $\Delta$ | Note                     |
|---------------|--------|----------|--------------------------|
| SF (baseline) | 0.3344 | —        | spreading_steps=0        |
| SF (top-10)   | 0.3454 | +0.0110  | reporting depth effect   |
| SF + QE       | 0.3261 | -0.0083  | Arabic synonym expansion |
| RRF k=60      | 0.3571 | +0.0117  | SF + BM25 fusion         |
| Dense Arabic  | 0.0030 | -0.3314  | multilingual embedding   |
| BM25          | 0.1550 | -0.1794  | lexical baseline         |



TABLE XVIII  
FAILING CONCEPT QUERIES WITH HYPOTHESIZED FAILURE CAUSES.

| Query                 | Key Terms                   | Failure Cause                           |
|-----------------------|-----------------------------|-----------------------------------------|
| Q011 judgment         | judgment, دين               | Polysemy (دين = religion/debt/judgment) |
| Q013 punishment hell  | punishment, hell, نار, عذاب | Terms too broad                         |
| Q014 mercy            | mercy, رحمه                 | Morphological variants                  |
| Q023 justice          | justice, قسط, عدل           | Synonyms not discriminative             |
| Q016 night power      | night, power                | “power” not in vocab                    |
| Q027 purpose humanity | purpose, humanity           | Generic terms                           |
| Q028 angels           | angel, ملكه                 | Vocabulary mismatch                     |
| Q029 hypocrites       | hypocrite, غافق             | Spreading noise                         |
| Q030 believers        | believer, مؤمن              | Fingerprint over-generalization         |

TABLE XIX  
SPREADING EFFECT ON FINGERPRINT SIZE ACROSS QUERY TYPES.

| Query Type               | Before Spread | After Spread | $\Delta$ |
|--------------------------|---------------|--------------|----------|
| Proper name (Joseph)     | 1,245         | 2,046        | +64%     |
| Proper name (Solomon)    | 1,803         | 2,788        | +55%     |
| Concept (prayer صلوه)    | 2,031         | 3,216        | +58%     |
| Concept (justice عدل)    | 2,013         | 3,082        | +53%     |
| Broad concept (believer) | 4,086         | 5,688        | +39%     |

TABLE XX  
FINAL RANKING OF ALL EVALUATED RETRIEVAL METHODS.

| Rank | Method          | MRR    | vs. SF |
|------|-----------------|--------|--------|
| 1    | RRF (SF + BM25) | 0.3571 | —      |
| 2    | SF (top-10)     | 0.3454 | —      |
| 3    | SF (baseline)   | 0.3344 | —      |
| 4    | BM25            | 0.1550 | −1.2×  |
| 5    | Dense (Arabic)  | 0.0030 | −115×  |
| 6    | Dense (English) | <0.001 | −345×  |

## REFERENCES

- [1] S. Robertson and H. Zaragoza, “The probabilistic relevance framework: BM25 and beyond,” *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.
- [2] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proc. EMNLP*, 2019.
- [3] F. Webber, “Semantic folding theory and its application in semantic fingerprinting,” 2018. [Online]. Available: <https://www.widdis.com/semantic-folding/>
- [4] L. van der Maaten and G. Hinton, “Visualizing data using t-SNE,” *Journal of Machine Learning Research*, vol. 9, pp. 2579–2605, 2008.
- [5] L. McInnes, J. Healy, and J. Melville, “UMAP: Uniform manifold approximation and projection for dimension reduction,” 2018, arXiv:1802.03426.
- [6] K. Taghva, R. Elkhoury, and J. Coombs, “Arabic stemming without a root dictionary,” in *Proc. ITCC*, 2005.
- [7] L. S. Larkey, L. Ballesteros, and M. E. Connell, “Improving stemming for Arabic information retrieval: Light stemming and co-occurrence analysis,” in *Proc. SIGIR*, 2002.
- [8] C. Zhai, “Statistical language models for information retrieval: A critical review,” *Foundations and Trends in Information Retrieval*, vol. 2, no. 3, pp. 137–213, 2008.
- [9] O. Benzaïa, D. Ziou, and A. Bouridane, “BM25 for Arabic information retrieval: A comparative study,” *Journal of Information Science*, vol. 46, no. 6, pp. 783–796, 2020.
- [10] M. Aly, J. Grönewald, and S. Mirrezaei, “Cross-lingual information retrieval: A survey,” *ACM Computing Surveys*, vol. 55, no. 2, pp. 1–35, 2022.
- [11] F. Webber, “Semantic folding in question answering systems,” *Journal of Artificial General Intelligence*, vol. 7, no. 1, pp. 1–25, 2016.